

UNIT 4: ALGEBRA

In algebra, we use letters to represent to represent unknown numbers.

4.1 Simplifying algebraic expressions

Put like terms together. A letter or a number with a letter in algebra is known as a term.

Example 1.

- i. $y + y + y + y + y$ is put together as $5 \times y = 5y$.
- ii. $m + m + m + m + m + m = 6 \times m = 6m$

When simplifying algebraic expressions, similar letters should be put together.

1. The operation sign (+ or -) placed before a term is for that term and determines the operation to be performed.
 - i. $+ 7n$ means add $7n$; $- 9a$ means subtract $9a$.
 - ii. $12a - 7a$ means subtract $7a$ from $12a$ which results to $5a$.
 $= 13y - 4y = 9y$
 - iii. $p + 2p + 3p - 4p = 2p$
 - iv. $3n + 7m + 15n - 4m - 11n$

Putting like terms together we get;

$$= (3n + 15n - 11n) + (7m - 4m) = 7n + 3m$$

$$= 2\frac{1}{2}p + 4\frac{1}{4}q - \frac{3}{4}p - 2\frac{1}{2}q$$

Putting like terms together

$$\begin{aligned} &= (2\frac{1}{2}p - \frac{3}{4}p) + (4\frac{1}{4}q - 2\frac{1}{2}q) \\ &= 1\frac{3}{4}p + 2q \end{aligned}$$

$$\text{v. } \frac{1}{2}(5x + 7y) + \frac{1}{4}(3x - 6y)$$

Multiply every term in the brackets by the value outside the bracket.

$$= \frac{5}{2}x + \frac{7}{2}y + 3/4x - 6/4y$$

Putting like terms together

$$= \frac{5}{2}x + \frac{3}{4}x + \frac{7}{2}y - \frac{3}{2}y = \frac{13}{4}x + \frac{4}{2}y$$

$$= \frac{13}{4}x + 2y$$

vi. $\frac{15a + 12b - 6a + 8b}{4(3a - 2b) + 13b}$

$$= \frac{15a - 6a + 12b + 8b}{12a - 8b + 13b}$$

$$= \frac{9a + 20b}{12a + 13b - 8b}$$

$$= \frac{9a + 20b}{12a + 5b}$$

Activity 1:

Simplify the algebraic expressions below:

i. $d + 3d - 5d + 7d$

ii. $\frac{1}{2}(18x + 24y) - \frac{1}{4}(20x + 16y)$

iii. $\frac{\frac{3}{5}(15a + 20b) - \frac{2}{3}(9a - 6b)}{\frac{3}{4}(8a - 12b) + 16b}$

iv. $2.5m + 7 - 1.45m + 13$

v. $\frac{5}{8}(32p - 16q) + \frac{3}{7}(14 + 21q)$

vi. $36(\frac{3}{4}x + \frac{5}{6}y - \frac{3}{8}x) - \frac{4}{5}(10x + 20y)$

vii. $(6 + 3)w + 11w - 6$

4.2 Word statements into algebraic expressions

We simplify algebraic expressions to find the unknown numbers.

Steps

- Identify the unknown quantity and give it a letter to represent it.
- Identify the operations to be used.

Example 2.

- A father is five times the age of his daughter. If the sum of their ages is 42 years, find the age of the daughter.

Solution

Let the age of the daughter be y .

$$\begin{aligned}y + 5y &= 42 \\6y &= 42 \\y &= \frac{42}{6} = 7\end{aligned}$$

Example 3.

Maryanne has n mangoes, Winnie has half Maryanne's mangoes while Debborah has twice Maryanne's mangoes. If they have 28 mangoes all together, how many mangoes does each have?

Solution.

Let the number of maryanne's mangoes be n .

Winnie has $\frac{n}{2}$ mangoes

Debborah has $2n$ mangoes

$$\begin{aligned}n + \frac{n}{2} + 2n &= 28 \\3n + \frac{n}{2} &= 28\end{aligned}$$

Multiplying both sides by 2

$$\begin{aligned}6n + n &= 56 \\7n &= 56 \\n &= \frac{56}{7} \quad n = 8\end{aligned}$$

Example 4.

The length of a rectangle is greater than its width by 12cm. If the perimeter of the rectangle is 64cm, find its length and area.

Solution.

$$P = 2(l + b)$$

Where $b = w$

$$l = w + 12$$

Let the width be w .

$$\text{Length} = w + 12$$

$$P = 2(l + b)$$

$$64 = 2(w + 12 + w)$$

$$64 = 2(2w + 12)$$

$$64 = 4w + 48$$

$$4w = 64 - 48$$

$$4w = 16$$

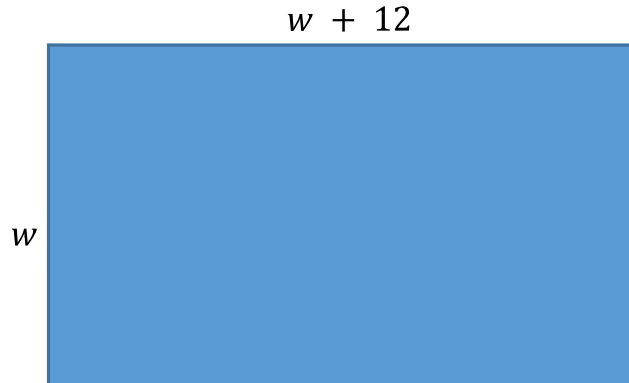
$$w = \frac{16}{4} \quad w = 4 \text{ cm}$$

$$\text{Length } (l) = w + 12$$

$$= 4 + 12 = \mathbf{16 \text{ cm}}$$

$$\text{Area} = l \times w$$

$$= 16 \times 4 = \mathbf{64 \text{ cm}^2}$$



Activity 2:

1. A mother is 6 times older than her son. The sum of their ages is 63 years, determine the age of the mother.
2. The perimeter of a square is 36 cm. Find the length of its sides.
3. David's age divided by 3 is equal to David's age minus 14. Find David's age.